Sampling and Variability

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for Sampling and Variability before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Tokens and Context

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Predictive and Generative AI

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Sampling and Variability trace and record the first step you can explain without help.

Use W2: Compute the adjusted table for **Sampling** and **temperature**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Sampling and Variability explanation against the supplied facts before

> discussing it.

Use the worked case for **Sampling** and **temperature**. Proposed explanation: "**Sampling** means equal chances". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Sample a new table independently

Probabilities A=0.5,B=0.3,C=0.2 in that order. Compute T=0.5 probabilities and cumulative boundaries. Trace u=0.5 and u=0.9 under T=1 and T=0.5. Explain the limit of the comparison.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Sampling and Variability target to a trace and a justified conclusion.

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Sampling and Variability case and show the deciding evidence.

One prompt produces three different valid quiz questions.

> Does **variability** mean failure?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 26

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Prompt A is run three times and returns three different stories. Does this alone prove the prompt changed? Describe a fair comparison of two prompts.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 4 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	

Test record	My evidence
Why the result occurred	

WORKBOOK · 9 OF 26

Project checkpoint

> Justify what Sampling and Variability adds to Verified Study Buddy and keep the evidence.

Use low **variability** for structured study answers and allow more diversity only for labeled brainstorming.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.

- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Trace boundaries

> Trace boundaries. Apply the supplied Sampling and Variability facts and keep your reasoning.

At T=1 select tokens for u=0,0.6,0.9 and 0.999. Explain the handling of u=1.

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Compute the adjusted table

Compute the adjusted table. Apply the supplied Sampling and Variability facts and keep your reasoning.

At T=0.5 compute squared weights, total and normalised probabilities. Check the sum.

My reasoning and evidence

WORKBOOK · 14 OF 26

W3 Compare a fixed draw

Compare a fixed draw. Apply the supplied Sampling and Variability facts and keep your reasoning.

Trace u=0.7 at T=1 and T=0.5. Does the different **output** demonstrate improved factual accuracy?

My reasoning and evidence

WORKBOOK · 15 OF 26

W4 Sample a new table independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Sample a new table independently. Apply the supplied Sampling and Variability facts and keep your reasoning.

Probabilities A=0.5,B=0.3,C=0.2 in that order. Compute T=0.5 probabilities and cumulative boundaries. Trace u=0.5 and u=0.9 under T=1 and T=0.5. Explain the limit of the comparison.

My reasoning and evidence

WORKBOOK · 16 OF 26

W5 Specify a valid experiment

> Specify a valid experiment. Apply the supplied Sampling and Variability facts and keep your reasoning.

A student changes the prompt and model as well as **temperature**, then attributes a better answer to **temperature**. Propose a fair comparison and reject any unsupported universal claim.

My reasoning and evidence

A small attention calculation — W1 Compute scores and weights

> Use the supplied Sampling and Variability task to record your reasoning, evidence and next step.

For q=[1], keys [0],[ln(3)] and d=1, compute **score**s and **softmax** weights.

My reasoning and evidence

A small attention calculation — W2 Combine and mask

> Use the supplied Sampling and Variability task to record your reasoning, evidence and next step.

Use values [2],[6] for the preceding weights. Compute **output** and **output** when the second key is excluded.

> **My reasoning and evidence**

A small attention calculation — W3 Trace equal scores

> Use the supplied Sampling and Variability task to record your reasoning, evidence and next step.

For q=[1,1], keys [1,0],[0,1], values [2,0],[0,4], compute scaled **score**s, weights and **output**.

> **My reasoning and evidence**

A small attention calculation — W4 Calculate new attention independently

> Use the supplied Sampling and Variability task to record your reasoning, evidence and next step.

q=[1], keys [0],[ln(2)],[ln(3)], values [0],[3],[8], width 1. Compute **score**s, weights and **output**. Then mask the third item and recompute. Explain whether **output** establishes a factual claim.

My reasoning and evidence

WORKBOOK · 21 OF 26

A small attention calculation — W5 Check stability and shape

> Use the supplied Sampling and Variability task to record your reasoning, evidence and next step.

Scores are 1000 and 1000+ln(3); values 4 and 8. Compute stable shifted weights and **output**. What if one key has width two and the query width one?

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

Pictorial case practice: Sampling and Variability

> A fictional two-token distribution illustrates variability in generation. Explain the labelled step

> before checking the changed condition.

A fictional two-token distribution illustrates **variability** in generation.

At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.

Use token order A,B and **cumulative interval**s [0,boundary),[boundary,1). Draw u=0.85.

AI ACADEMY / AI-3/4

Temperature-scaled toy probabilities

	T=1	T=0.5
A	0.8	0.9411764705882353
B	0.2	0.058823529411764705

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/4

Sampling and Variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Original distribution

A boundary 0.8; u=0.85 selects B.

2. T=0.5 weights

Square probabilities:0.64 and 0.04; total 0.68.

3. Normalise

A=16/17≈0.9412; B=1/17≈0.0588.

4. Same draw

0.85 is below 16/17, so T=0.5 selects A.

CHANGE ONE THING

Set the draw to 0.97, keeping both distributions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional two-token distribution illustrates **variability** in generation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? 0.85 is below 16/17, so T=0.5 selects A.

> 1. Normalise

2. Original distribution

3. T=0.5 weights

4. Same draw

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Same draw 0.85 is below 16/17, so T=0.5 selects A. Lower positive **temperature** concentrates this unequal distribution on its larger entry.

It changes **sampling** probabilities, not **evidence** for whether A or B is true. Using the same draw isolates how the distribution changes the decision.

This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Set the draw to 0.97, keeping both distributions. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Both **temperature**s select B.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both **temperature**s select B.

> 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Repair a missing building block

Review Tokens and Context (ai-3/3). Goal: Account for fixed text, **evidence** and reserved **output** under a toy budget. Short practice: Can A alone justify answering 20 seats after the exception is known?.

Return to this week and explain what you changed.

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Sampling and Variability picture from its labels, then record a relevant limit.

01. Sampling and Variability: the case

AI ACADEMY / AI-3/4 · 01 · LOOK AT THE FACTS

Sampling and Variability: the case

The situation

A fictional two-token distribution illustrates variability in generation.

Supplied evidence

At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.
Use token order A,B and cumulative intervals [0,boundary),[boundary,1). Draw u=0.85.

By the end, I can calculate a toy temperature change and trace a sampling draw.

Follow the labelled picture with the matching explanation. Open the vector diagram

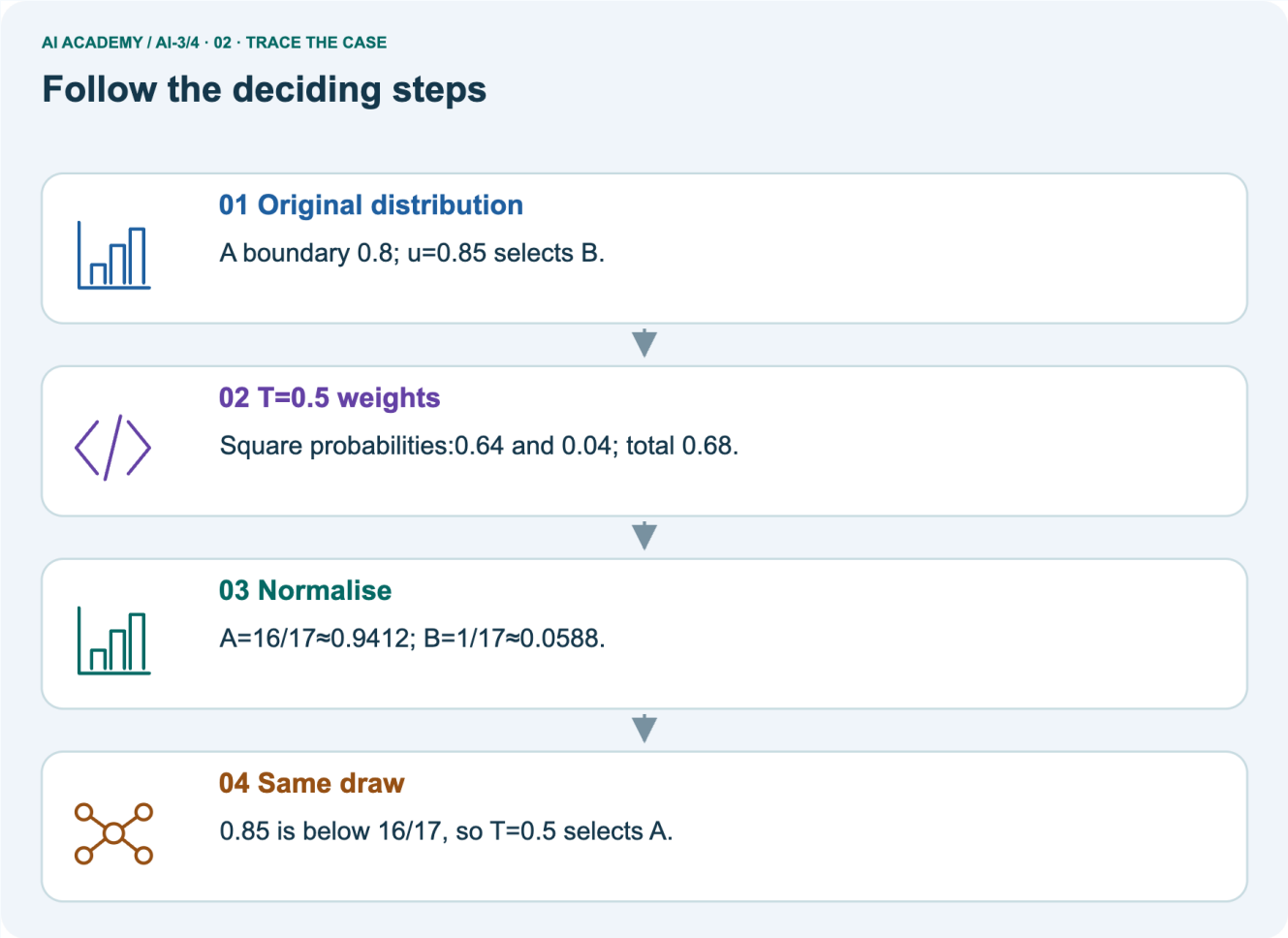
A fictional two-token distribution illustrates **variability** in generation. Supplied facts: At T=1, A=0.8 and B=0.2. For positive T use p^(1/T), then normalise.

Use token order A,B and **cumulative interval**s [0,boundary),[boundary,1). Draw u=0.85.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



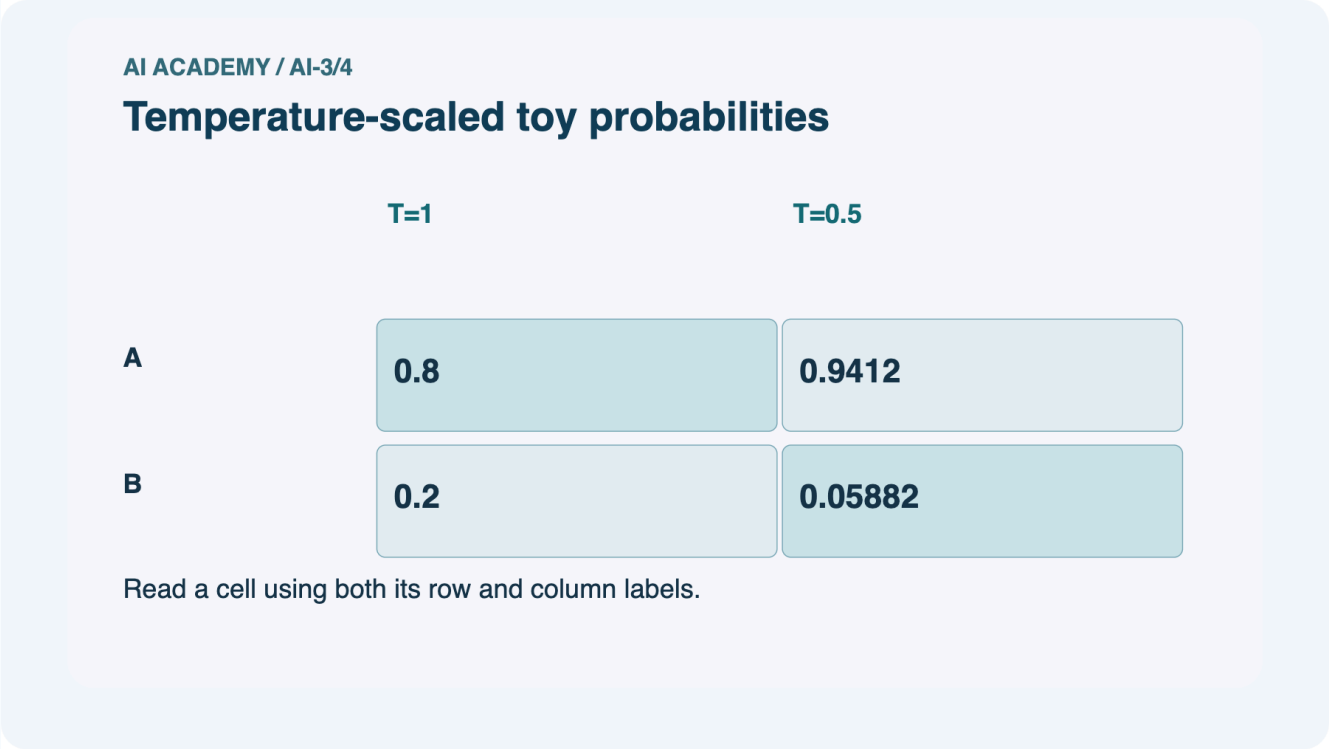
Follow the labelled picture with the matching explanation. Open the vector diagram

Original distribution: A boundary 0.8; u=0.85 selects B. → T=0.5 weights: Square probabilities:0.64 and 0.04; total 0.68. → Normalise: A=16/17≈0.9412; B=1/17≈0.0588. → Same draw: 0.85 is below 16/17, so T=0.5 selects A.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Temperature-scaled toy probabilities



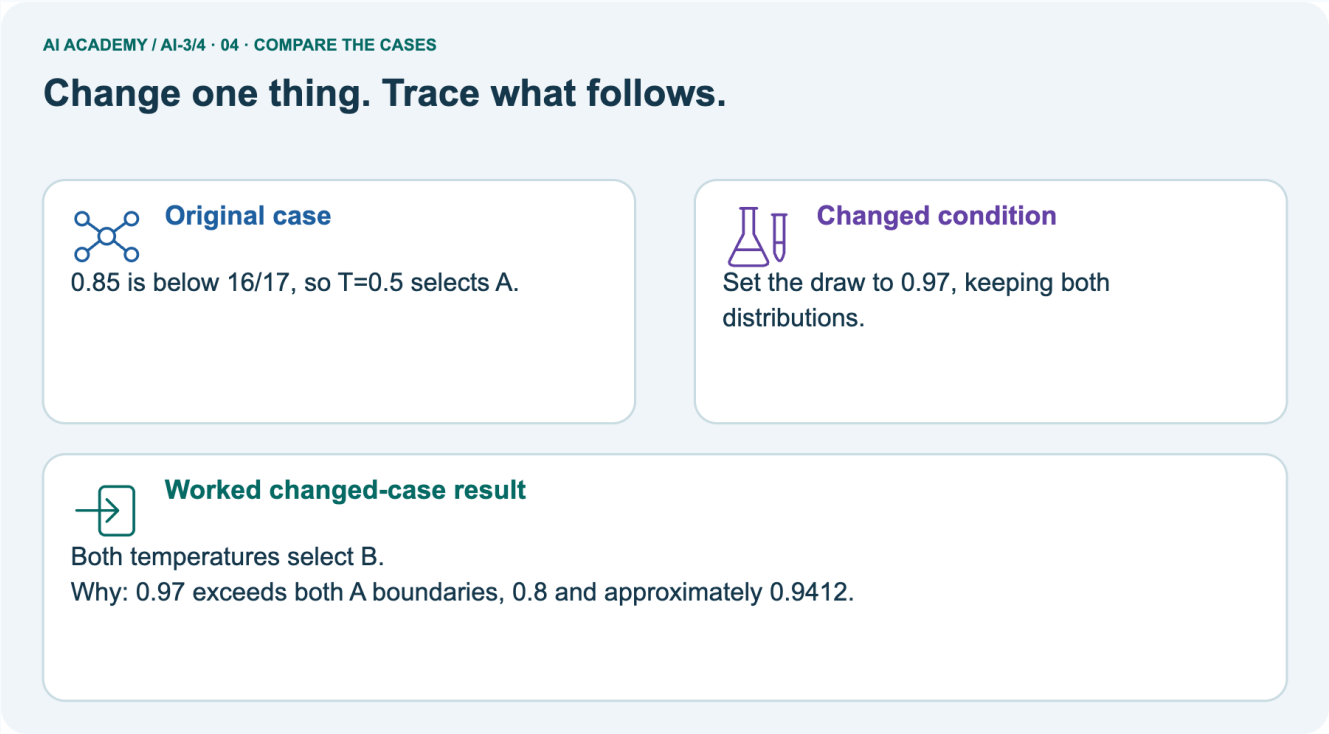
Follow the labelled picture with the matching explanation. Open the vector diagram

Temperature-scaled toy probabilities. A: T=1 = 0.8, T=0.5 = 0.9411764705882353. B: T=1 = 0.2, T=0.5 = 0.058823529411764705.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 0.85 is below 16/17, so T=0.5 selects A. Changed condition: Set the draw to 0.97, keeping both distributions. Expected result: Both **temperature**s select B.

Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit

AI ACADEMY / AI-3/4 · 05 · EXPLAIN WITH CARE

Connect the explanation to its limit

What the case explains

Lower positive temperature concentrates this unequal distribution on its larger entry. It changes sampling probabilities, not evidence for whether A or B is true. Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a temperature-scaled model distribution.

What it does not establish

Two tokens and one draw do not predict every output; temperature alone does not solve hallucinations.

Follow the labelled picture with the matching explanation. Open the vector diagram

Lower positive **temperature** concentrates this unequal distribution on its larger entry. It changes **sampling** probabilities, not **evidence** for whether A or B is true.

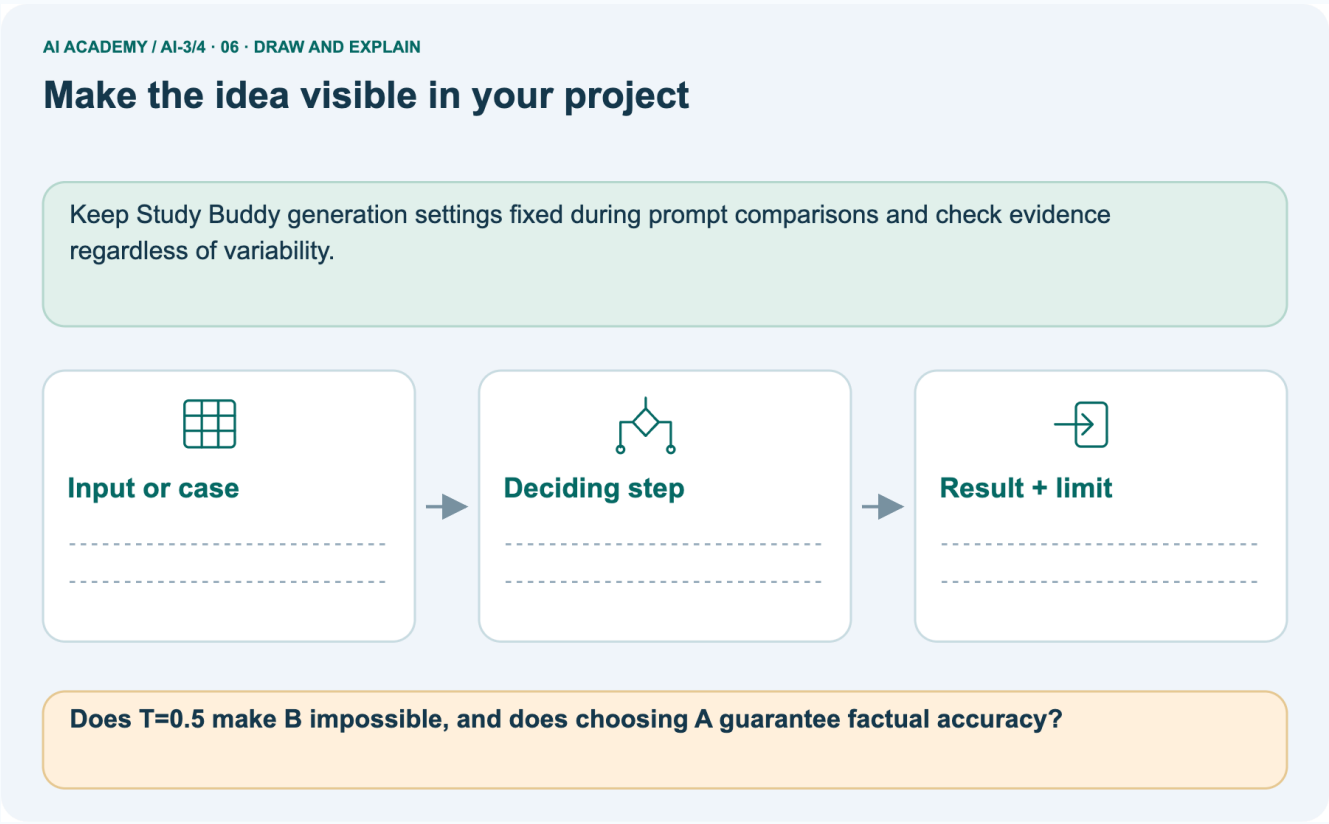
Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a **temperature**-scaled model distribution.

Limit: Two tokens and one draw do not predict every **output**; **temperature** alone does not solve hallucinations.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep Study Buddy generation settings fixed during prompt comparisons and check **evidence** regardless of **variability**. Draw your **input**, deciding step, result and limit.

Supported question: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.